

Gym Membership Churn Analysis

Hypothesis Testing & Classification Modeling on the Model Fitness Dataset

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1. Executive Summary

This report presents a churn analysis for a fitness club membership base, combining statistical hypothesis testing with a logistic regression classification model to identify which member segments are most likely to cancel their membership and what factors drive that risk.

Using data from 4,000 members, the analysis finds an overall churn rate of 26.5% and identifies a set of statistically significant drivers of churn: a drop in class-attendance frequency relative to a member's own historical average, lack of group-class participation, contract length, membership tenure ("lifetime"), and age group. The resulting classification model distinguishes churned from retained members with high accuracy (ROC-AUC of 0.97). These statistical findings are paired with domain-informed interpretation, drawing on ten years of hands-on fitness and BJJ coaching experience, to translate the numbers into actionable retention recommendations for gym management.

2. Business Context

Member churn is one of the most consequential and controllable metrics for a fitness business: acquiring a new member is substantially more expensive than retaining an existing one, and even small improvements in retention compound over a membership base's lifetime. This project treats churn analysis as a business problem, structured around one main question and a set of supporting sub-questions.

Main question: Which member segments carry the highest churn risk, and what factors drive that risk?

Sub-questions:

1. Does a drop in a member's class-attendance frequency, relative to their own historical average, predict churn?
2. Does participation in group classes reduce churn risk?
3. How does contract length relate to churn?
4. Do convenience and affiliation factors — proximity to the gym, corporate/partner membership, signing up through a friend's referral — affect retention?
5. How does churn risk vary by member age and gender?

The analysis proceeds in two stages. Hypothesis testing is used first to determine which candidate variables have a statistically meaningful relationship with churn. A logistic regression model is then built on the validated feature set, both to quantify each factor's independent contribution once the others are controlled for, and to classify which members are at risk of churning.

3. Data Overview & Limitations

The analysis uses the "Model Fitness" dataset: 4,000 members and 14 original features, covering membership terms (contract length, months remaining, partner/corporate affiliation, friend referral, proximity to the gym), engagement (group-class participation, total vs. current-month class-attendance frequency), demographics (age, gender), spend (average additional charges), and the churn label itself. The dataset is fully synthetic and was quality-checked at the start of the project: no missing values and no duplicate records.

This dataset is synthetic. The specific figures, coefficients, and percentages reported throughout this analysis are internally consistent within this dataset but should not be read as empirical claims about real-world gym populations. Where this report draws on real coaching experience to interpret a pattern, that interpretation is clearly separated from the statistical findings (see Section 6) and should be read as a plausibility check, not independent proof.

Several limitations are worth noting:

- **Measurement validity of "churn."** The churn label captures whether a member left this gym, not whether they gave up on fitness altogether. This matters most for the age-group finding (Section 6): younger members' higher churn may partly reflect ordinary life-stage mobility (changing schools, jobs, or cities) rather than lower commitment.
- **No time/seasonality data.** The dataset has no sign-up date field, so it cannot test whether membership timing (e.g., a seasonal summer sign-up) affects churn.
- **No gym occupancy data.** Crowding at the time of sign-up, a plausible retention factor, is not captured.
- **No athletic/sports-history data.** A member's prior exercise background, which could moderate several of the effects discussed below, is not available.

These gaps are treated as scope boundaries rather than flaws: they define what this analysis can and cannot speak to, and are revisited as future-work opportunities in Section 7.

4. Methodology

Exploratory Data Analysis

The overall churn rate is 26.5%. A new feature, `frequency_drop`, was engineered as the difference between a member's historical and current-month class-attendance frequency (total – current month), so that a positive value means attendance slowed down and a negative value means it picked up. Churn rates were also profiled across every categorical variable (gender, partner, friend referral, group classes, proximity, contract length, age group) and against tenure and spend.

Hypothesis Testing

Two types of statistical tests were used to validate which variables have a statistically meaningful relationship with churn before including them in a predictive model:

- A Mann-Whitney U test confirmed that `frequency_drop` is significantly higher among churned members than retained members ($U = 553,037$, $p \approx 4.9 \times 10^{-214}$).
- Chi-square tests of independence, summarized by Cramér's V (≈ 0.1 weak, ≈ 0.3 considerable, ≈ 0.5 strong), were run for each categorical variable against churn:

Variable	χ^2	p-value	Cramér's V
Contract Period	653.56	1.21×10^{-142}	0.404
Age Group	417.40	2.31×10^{-91}	0.323
Group Classes	122.15	2.14×10^{-28}	0.175
Friend Discount	104.48	1.58×10^{-24}	0.162
Partner Company	99.12	2.37×10^{-23}	0.157
Near Location	64.84	8.14×10^{-16}	0.127
Phone Number	~ 0.00	0.989	0.0002
Gender	~ 0.00	0.993	0.0001

Gender and phone number showed no meaningful association with churn and were dropped from further analysis.

Feature Engineering

Beyond gender and phone, raw age was dropped in favor of the categorical age group, and month_to_end_contract and the raw attendance-frequency columns were dropped as redundant with contract_period and frequency_drop, respectively (confirmed via a correlation heatmap). The remaining categorical variables (contract_period, age_group) were one-hot encoded, and a Variance Inflation Factor (VIF) check was run on the full encoded feature set to rule out multicollinearity — all VIFs came back well within acceptable bounds (maximum ≈ 1.36 , excluding the intercept term), confirming the features are safe to model together.

Model Choice

statsmodels.Logit was used instead of scikit-learn's LogisticRegression because the project's goal is explanatory as well as predictive: statsmodels reports coefficient-level significance (p-values, confidence intervals), which is what makes it possible to say which factors matter and how confidently, not just to classify members. The data was split 80/20 into training and test sets, stratified on churn to preserve the 26.5% churn rate in both (train: 3,200 rows, 26.53% churn; test: 800 rows, 26.5% churn).

5. Model & Evaluation

The fitted model is highly significant overall (Pseudo $R^2 = 0.656$, LLR p-value < 0.001). The following variables were statistically significant ($p < 0.05$) predictors of churn, holding the others constant:

Variable	Odds Ratio	Interpretation
frequency_drop	≈ 1.59 / +0.1 unit	Strongest, most direct behavioral predictor of churn
lifetime (tenure)	0.354	Each extra month of membership lowers churn odds substantially
contract_period = 12mo	0.053 vs. 1mo	Longer contracts strongly associated with lower churn
contract_period = 6mo	0.214 vs. 1mo	Same direction, smaller effect
age_group = 36+	0.007 vs. 18–25	Much lower odds; wide confidence interval (see note below)
age_group = 26–35	0.152 vs. 18–25	Lower odds than youngest group
group_visits	0.448	Group-class participation roughly halves churn odds
avg_additional_charges_total	0.995 / \$1	Small but consistent: higher spend, lower churn
promo_friends	0.650 ($p=0.024$)	Marginal; much weaker once other variables are controlled for

near_location and partner were significant on their own in the hypothesis-testing stage but were **not** significant in the full model ($p = 0.590$ and $p = 0.814$ respectively). This is a standard and expected pattern: a variable can correlate with churn in isolation while carrying little additional information once correlated variables (such as contract length or group-class participation) are already in the model.

A note on quasi-separation

The model reported: “Possibly complete quasi-separation: a fraction 0.12 of observations can be perfectly predicted.” This is a common, soft form of separation — the model still converges normally, but a few individual coefficient estimates (most notably `age_group_36+`, which has an unusually wide confidence interval) are less numerically stable. It is treated here as a caveat on the precision of those specific estimates, not as a flaw that invalidates the model or its predictive performance.

Evaluation on the held-out test set

(800 members, unseen during training)

Metric	Retained (0)	Churned (1)
Precision	0.94	0.90
Recall	0.97	0.83
F1-score	0.95	0.86

Overall accuracy: 93%. ROC-AUC: **0.972** — the model reliably ranks churned members as higher-risk than retained members across almost any decision threshold, indicating strong, well-generalizing discriminative performance rather than an artifact of class imbalance.

6. Findings

Findings are presented in two clearly separated layers, as discussed in Section 3: statistical results specific to this synthetic dataset, and a separate layer of domain interpretation drawing on ten years of hands-on fitness and BJJ coaching experience. The second layer is a plausibility check on why a pattern might exist, not an empirical claim.

Statistical findings (this dataset)

`frequency_drop` is the single strongest predictor of churn. Group-class participation, membership tenure, and longer contract length are all strongly protective against churn. Older age groups churn far less than 18–25 year-olds. Proximity, corporate/partner affiliation, friend referral, gender, and phone number show weak to no independent effect once other factors are accounted for.

Domain interpretation, by variable

- **frequency_drop.** A drop in attendance frequency plausibly reflects a mix of familiar causes: expectations set at sign-up that don't match reality (often shaped by outside marketing or social media rather than by the gym itself), an onboarding pace that is too intense and leads to early burnout, and monotony — which group classes and coach attention can help offset.
- **group_visits.** The retention effect likely runs through reduced monotony and social connection — meeting people and working toward something together is a well-established driver of group-class adherence. For members who prefer not to train in a group setting, or for smaller gyms without the capacity to run group classes, a strong, attentive coach relationship is a comparable alternative channel — not merely a fallback, but a route that reaches a different member segment.
- **lifetime.** The “sunk cost” explanation (“we already paid, don't want to waste it”) is judged to be a weak driver here, since it is a one-time lump payment whose psychological pull tends to fade over the multi-month spans involved. Habit formation is a more plausible driver: continued regular attendance likely

reduces the desire to quit over time, though whether this reflects a distinct early “threshold” or a gradual, individual-dependent process is not something this dataset can resolve on its own.

- **contract_period.** The association between longer contracts and lower churn is interpreted primarily as a selection effect — members who choose longer, cheaper-per-month contracts are likely already more confident about sticking with the gym — rather than the contract itself causing loyalty.
- **age_group.** Older members' more routine-set lives plausibly support attendance continuity, while younger members change gyms more often simply due to school or job transitions. This connects to a genuine measurement-validity caveat (Section 3): this dataset's churn label may conflate “left this gym” with “gave up on fitness,” so younger members' higher churn may partly reflect life-stage mobility rather than lower commitment. The relationship is not purely one-directional, either — some younger members show longer continuation by channeling energy into training.
- **promo_friends.** The retention mechanism behind a friend referral is likely the same reduced-monotony and social-connection effect seen for group_visits, rather than a direct effect of having started with a friend. This is consistent with the model: promo_friends' effect became only marginally significant once other variables were controlled for.

7. Business Recommendations & Conclusion

Recommendations

1. **Monitor attendance frequency as an early-warning signal.** Track each member's frequency_drop and trigger proactive outreach once it crosses a meaningful threshold. At sign-up, set clear expectations (not just “honest” ones) about realistic progress timelines, framing fitness as a long-term routine rather than a short-term goal — mismatched expectations are often shaped by outside influences, not by gym messaging itself.
2. **Use group classes, and coach attention, to counter monotony.** Pair members flagged by a frequency drop with group classes. Where group-class capacity is limited, or for members who prefer not to train in a group, invest in a strong, attentive coach relationship as an alternative retention channel that reaches a different member segment.
3. **Concentrate retention effort in the first few months.** Early tenure is the highest-risk window; early engagement has an outsized effect on lifetime retention.
4. **Don't sell long contracts as a churn-reduction lever on their own.** The contract-length effect looks more like self-selection than causation; instead, target short/monthly-contract members for extra engagement.
5. **Separate “switched gyms” from real disengagement.** Add an exit-survey or reason field (or capture this context at registration) to address the age-group measurement-validity limitation directly.
6. **No standalone action needed for friend referrals** — their retention effect is already captured indirectly through the group-class and coach-attention recommendations above.

Conclusion

Combining hypothesis testing with a logistic regression classification model produced a well-evaluated (ROC-AUC 0.97) and internally consistent picture of churn drivers in this dataset, paired with coaching-informed interpretation to make the findings actionable. The clearest next steps for extending this work are to repeat the analysis on a real-world dataset — ideally one with enrollment dates to test the seasonality question raised but untestable here — and, where available, to incorporate gym occupancy at sign-up and members' athletic history, both flagged early in the project as relevant but unavailable in this dataset.